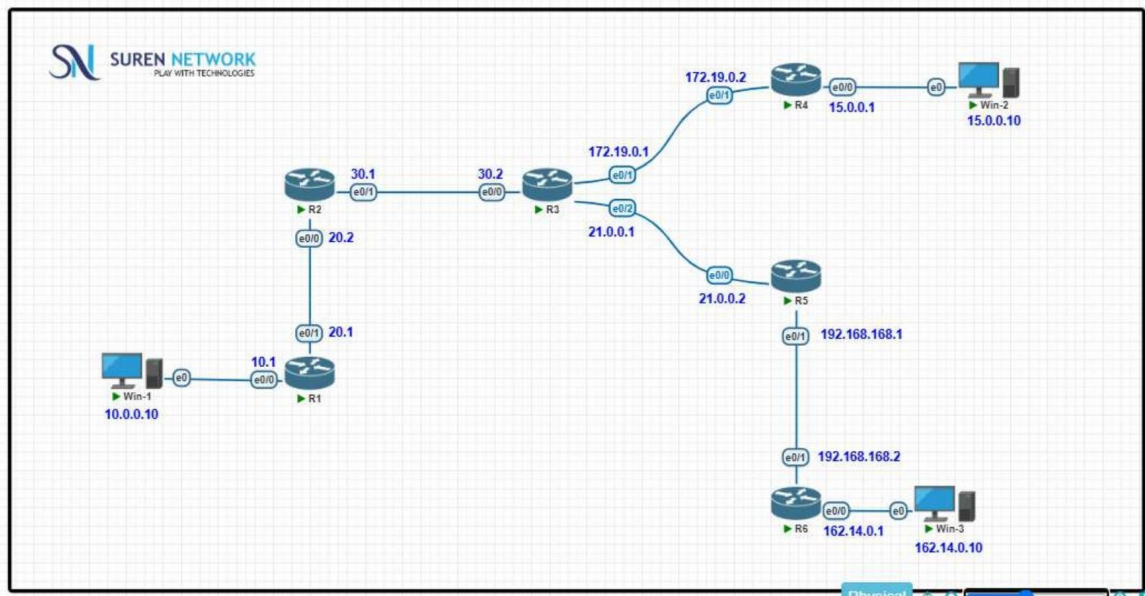


CCNA Practical Lab Report

Static Routing — 6 Routers, 3 Subnets, Full Connectivity

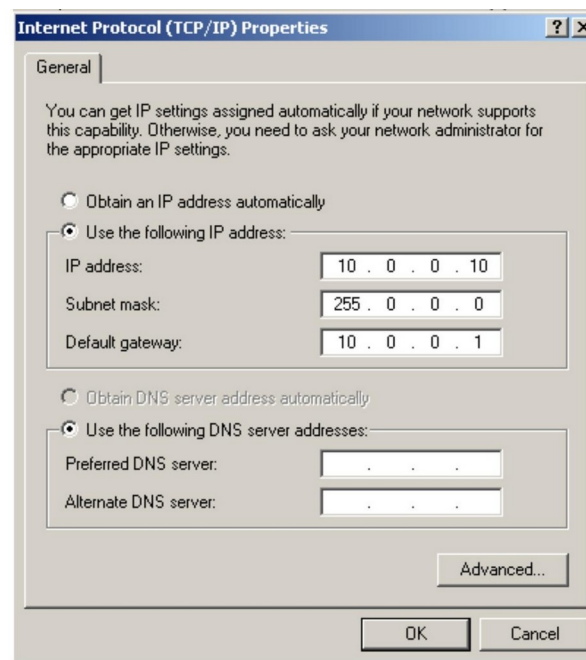
Six routers (R1–R6) were addressed and statically routed to connect three hosts (Win-1, Win-2, Win-3) across seven subnets. All configurations and ping tests below confirm full, loss-free connectivity.

Topology

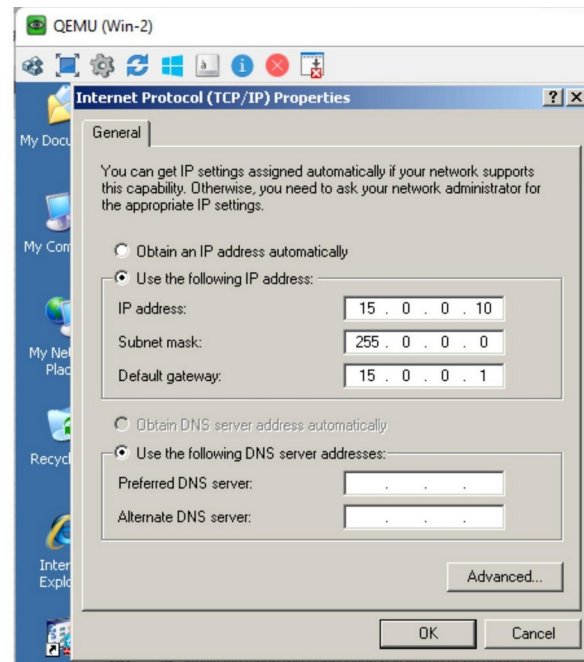


PC IP Configuration

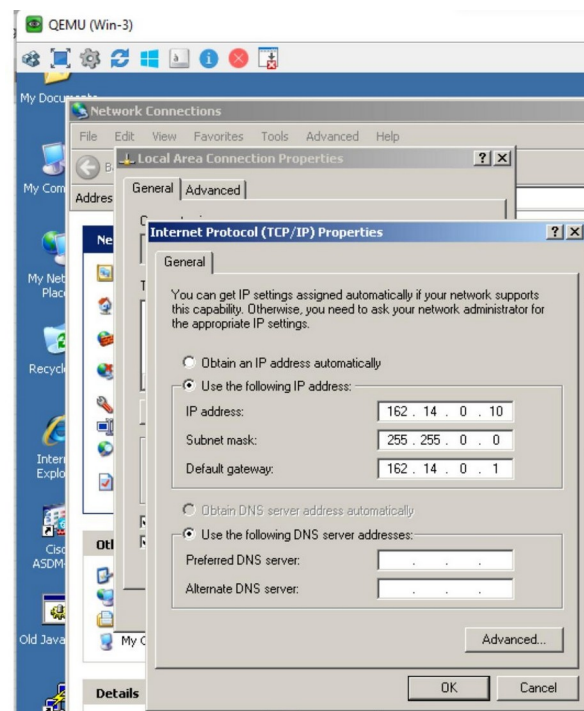
Win-1 — 10.0.0.10



Win-2 — 15.0.0.10



Win-3 — 162.14.0.10



Router Interface & Routing Tables

R1

```

R1#
R1#sh ip int br
Interface                IP-Address    OK? Method Status    Protocol
Ethernet0/0              10.0.0.1     YES manual up        up
Ethernet0/1              20.0.0.1     YES manual up        up
Ethernet0/2              unassigned   YES TFTP  down      down
Ethernet0/3              unassigned   YES TFTP  down      down
R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
R1#

```

R2

```

R2#
R2#sh ip int br
Interface                IP-Address    OK? Method Status    Prot
ocol
Ethernet0/0              20.0.0.2     YES manual up        up
Ethernet0/1              30.0.0.1     YES manual up        up
Ethernet0/2              unassigned   YES TFTP  down      down
Ethernet0/3              unassigned   YES TFTP  down      down
R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S       10.0.0.0/8 [1/0] via 20.0.0.1
S       15.0.0.0/8 [1/0] via 30.0.0.2
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.2/32 is directly connected, Ethernet0/0
S       21.0.0.0/8 [1/0] via 30.0.0.2
C       30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.1/32 is directly connected, Ethernet0/1
S       162.14.0.0/16 [1/0] via 30.0.0.2
S       172.19.0.0/16 [1/0] via 30.0.0.2
S       192.168.168.0/24 [1/0] via 30.0.0.2
R2#

```

R3

```

R3#
R3#sh ip int br
Interface                               IP-Address    OK? Method Status    Prot
Ethernet0/0                             30.0.0.2      YES manual up        up
Ethernet0/1                             172.19.0.1    YES manual up        up
Ethernet0/2                             21.0.0.1      YES manual up        up
Ethernet0/3                             unassigned    YES TFTP  down        down

R3#
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
C    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
     C    21.0.0.0/8 is directly connected, Ethernet0/2
     L    21.0.0.1/32 is directly connected, Ethernet0/2
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
     C    30.0.0.0/8 is directly connected, Ethernet0/0
     L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
     C    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
     L    172.19.0.0/16 is directly connected, Ethernet0/1
     L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
R3#

```

R4

```

R4#
R4#sh ip int br
Interface                               IP-Address    OK? Method Status    Prot
Ethernet0/0                             15.0.0.1      YES manual up        up
Ethernet0/1                             172.19.0.2    YES manual up        up
Ethernet0/2                             unassigned    YES TFTP  down        down
Ethernet0/3                             unassigned    YES TFTP  down        down

R4#
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
C    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
     L    15.0.0.0/8 is directly connected, Ethernet0/0
     L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
     C    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
     L    172.19.0.0/16 is directly connected, Ethernet0/1
     L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4#

```

R5

```

R5#
R5#sh ip int br
Interface                IP-Address      OK? Method Status      Prot
Ethernet0/0              21.0.0.2        YES manual up          up
Ethernet0/1              192.168.168.1   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down         down
Ethernet0/3              unassigned      YES TFTP  down         down

R5#
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/0
L    21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.1/32 is directly connected, Ethernet0/1
R5#

```

R6

```

R6#
R6#sh ip int br
Interface                IP-Address      OK? Method Status      Prot
Ethernet0/0              162.14.0.1      YES manual up          up
Ethernet0/1              192.168.168.2   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down         down
Ethernet0/3              unassigned      YES TFTP  down         down

R6#
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
S    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    162.14.0.0/16 is directly connected, Ethernet0/0
L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.2/32 is directly connected, Ethernet0/1
R6#

```

Connectivity Verification (Ping Tests)

From Win-1 (10.0.0.10)


```

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\User>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\User>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>

```

From Win-2 (15.0.0.10)

```

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 15.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 15.0.0.1

C:\Documents and Settings\User>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\User>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

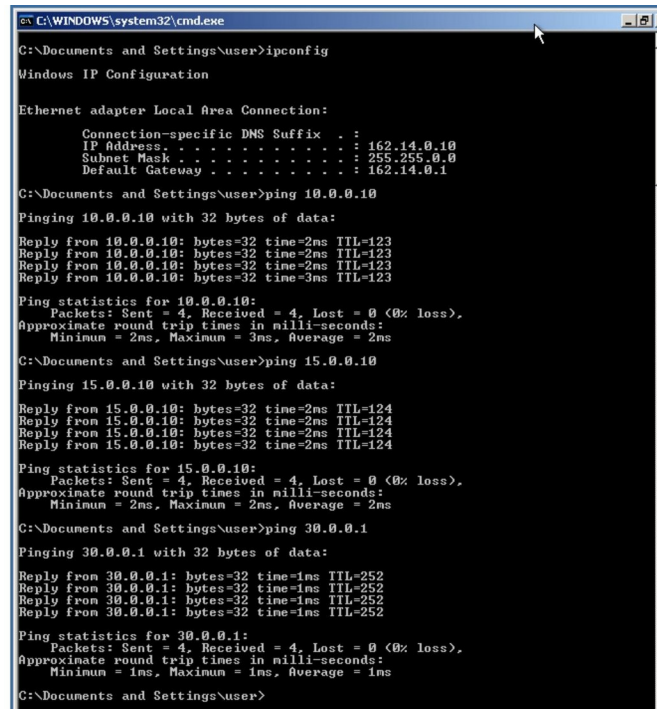
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>

```

From Win-3 (162.14.0.10)



```

C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 162.14.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 162.14.0.1

C:\Documents and Settings\User>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\User>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time=1ms TTL=252
Reply from 30.0.0.1: bytes=32 time=1ms TTL=252
Reply from 30.0.0.1: bytes=32 time=1ms TTL=252
Reply from 30.0.0.1: bytes=32 time=1ms TTL=252

Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\User>

```

Result

All ping tests between Win-1, Win-2, and Win-3 succeeded with 0% packet loss, confirming correct static routing across all six routers.